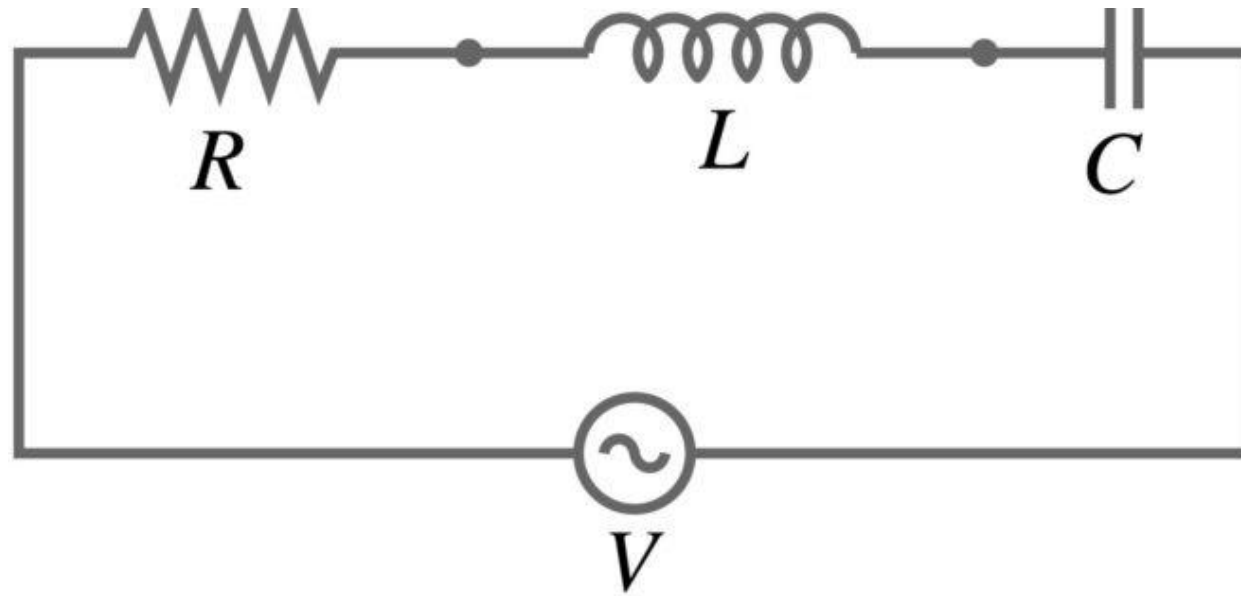




MODELING A SERIES RLC CIRCUIT

KRISHANGA

CMSE201-008

MODEL

How do changes in resistance (R), capacitance (C) and inductance (L) impact the transient and steady-state behavior of a series RLC circuit under varying alternating current (AC) conditions?

- Frequency
- Resistance
- Capacitance
- Inductance

$$\frac{di_1}{dt} = i_2$$

$$\frac{di_2}{dt} = \frac{V(t) - Ri_2 - \frac{1}{C}i_1}{L}$$

$$L \frac{d^2i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{1}{C}i(t) = V(t)$$

```
# Differential equation for series RLC circuit: L * d^2i/dt^2 + R * di/dt + (1/C) * i = V(t)
def rlc_system(t, y):
    i1, i2 = y # i1 is current i(t), i2 is the derivative di/dt
    V = V_amplitude * np.sin(omega * t) # AC voltage source V(t)
    di1_dt = i2
    di2_dt = (V - R * i2 - (1 / C) * i1) / L
    return [di1_dt, di2_dt]
```

```

function to solve and plot for different parameters
def rlc_response(R, L, C, f, V_amplitude=1):
    omega = 2 * np.pi * f # Angular frequency in rad/s

    def rlc_system(t, y):
        i1, i2 = y # i1 is current i(t), i2 is the derivative di/dt
        V = V_amplitude * np.sin(omega * t) # AC voltage source V(t)
        di1_dt = i2
        di2_dt = (V - R * i2 - (1 / C) * i1) / L
        return [di1_dt, di2_dt]

    # Initial conditions: i(0) = 0 (no initial current), di/dt = 0 (no initial rate of change of current)
    initial_conditions = [0, 0]

    # Time vector: simulate for 10 ms
    t_span = (0, 0.01)
    t_eval = np.linspace(0, 0.01, 1000)

    # Solve the differential equation using solve_ivp
    solution = solve_ivp(rlc_system, t_span, initial_conditions, t_eval=t_eval)

    # Extract current and its derivative
    current = solution.y[0] # i(t)
    di_dt = solution.y[1] # di/dt

    # Voltage across components
    voltage_R = R * current # Voltage across the resistor
    voltage_L = L * di_dt # Voltage across the inductor
    voltage_C = np.cumsum(current) / len(current) / C # Approximate voltage across the capacitor (du/dt = i/C)

    # Applied AC voltage
    voltage_source = V_amplitude * np.sin(omega * t_eval)

    # Plot the current, voltage across components, and input voltage
    plt.figure(figsize=(10, 8))

    # Plot voltage across resistor
    plt.subplot(4, 1, 1)
    plt.plot(t_eval, voltage_R, label=f'Voltage across Resistor (R={R} Ω)', color='blue')
    plt.title('Voltage Across Resistor, Inductor, and Capacitor')
    plt.xlabel('Time (s)')
    plt.ylabel('Voltage (V)')
    plt.grid(True)
    plt.legend()

    # Plot voltage across inductor
    plt.subplot(4, 1, 2)
    plt.plot(t_eval, voltage_L, label=f'Voltage across Inductor (L={L} H)', color='green')
    plt.xlabel('Time (s)')
    plt.ylabel('Voltage (V)')
    plt.grid(True)
    plt.legend()

    # Plot voltage across capacitor
    plt.subplot(4, 1, 3)
    plt.plot(t_eval, voltage_C, label=f'Voltage across Capacitor (C={C} F)', color='red')
    plt.xlabel('Time (s)')
    plt.ylabel('Voltage (V)')
    plt.grid(True)
    plt.legend()

    # Plot input AC voltage
    plt.subplot(4, 1, 4)
    plt.plot(t_eval, voltage_source, label=f'Input Voltage (f={f} Hz)', color='purple')
    plt.xlabel('Time (s)')
    plt.ylabel('Voltage (V)')
    plt.grid(True)
    plt.legend()

    plt.tight_layout()
    plt.show()

```

MODEL

- A function was created to reduce tedious code
- Solve_ivp is used to solve the differential equation.
- The base values that will be are:

$$f = 500 \text{ Hz}$$

$$R = 5 \text{ } \Omega$$

$$L = 1 \text{ mH}$$

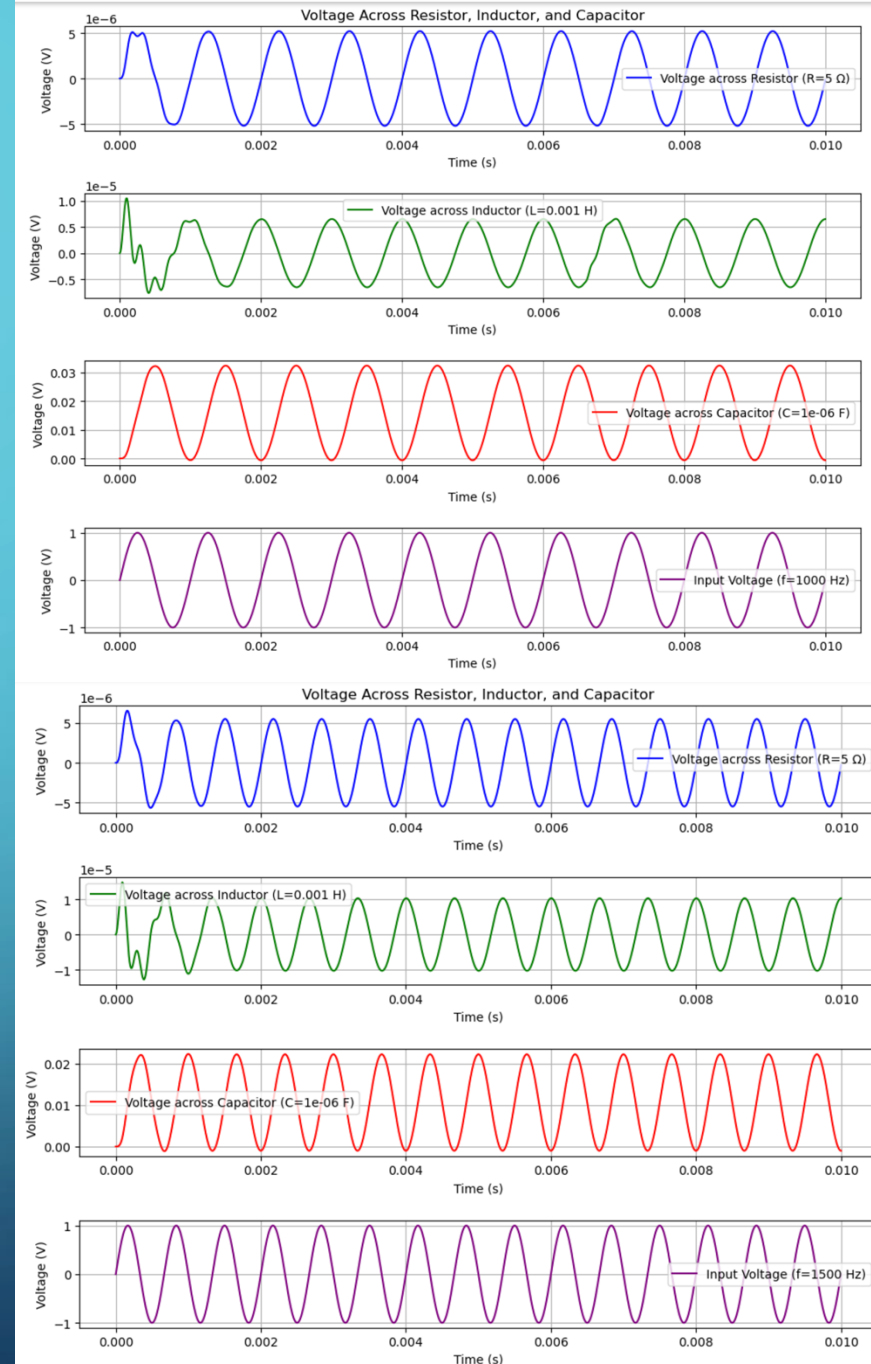
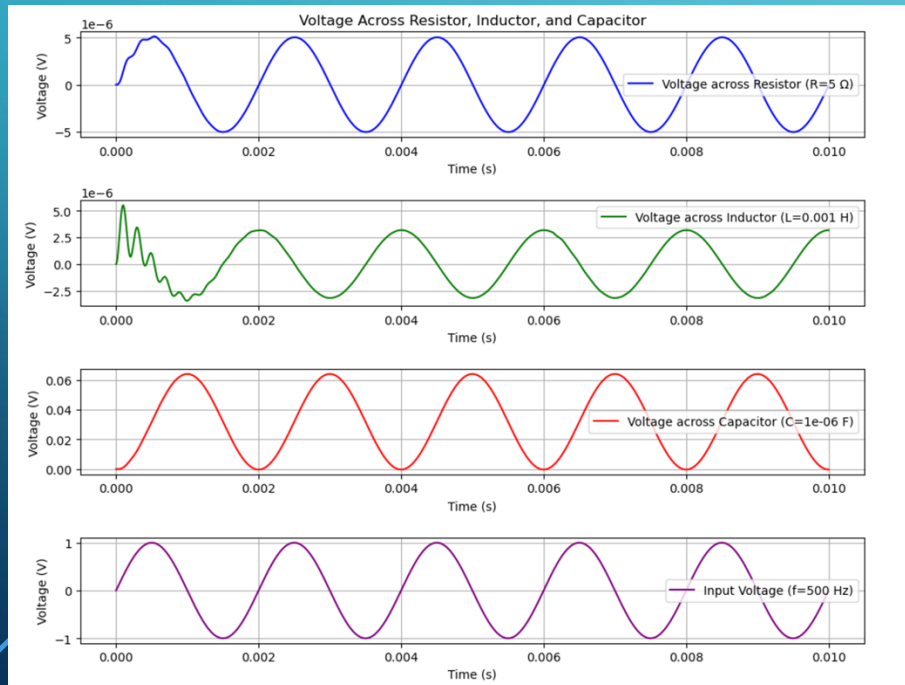
$$C = 1 \text{ } \mu\text{F}$$

VARYING FREQUENCY

Significant distortions in the inductor

```
#Varying frequencies
frequencies = [500, 1000, 1500] # Varying frequencies in Hz
resistances = [5]
inductances = [1e-3]
capacitances = [1e-6]

# Loop through different parameter sets and plot the results
for f in frequencies:
    for R in resistances:
        for L in inductances:
            for C in capacitances:
                rlc_response(R, L, C, f)
```

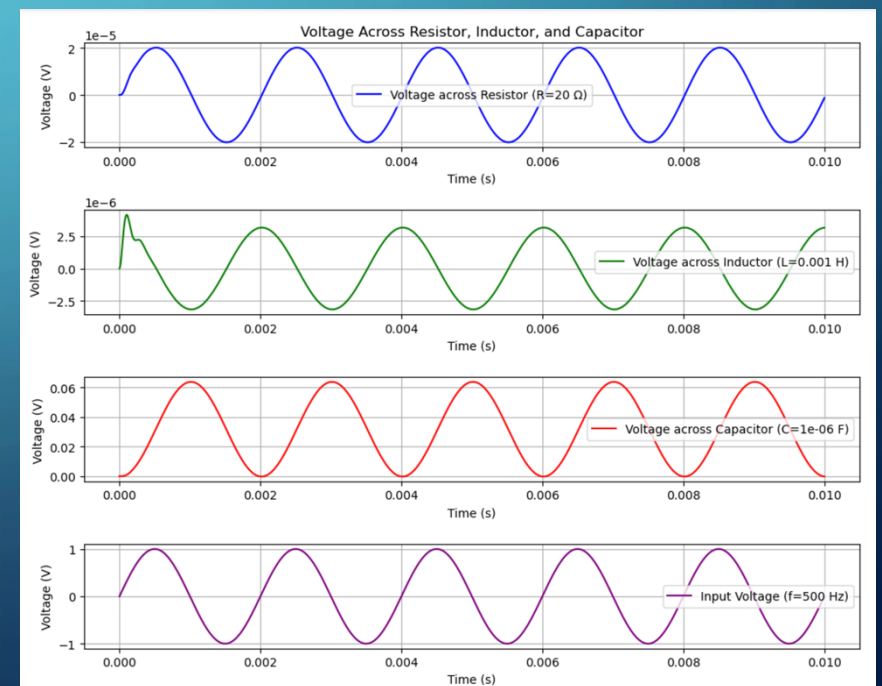
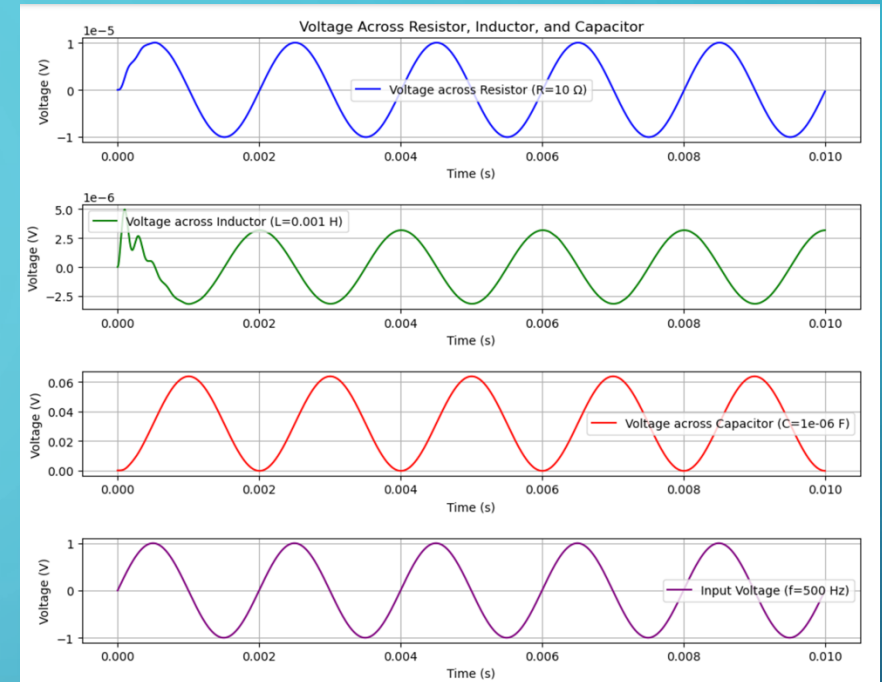
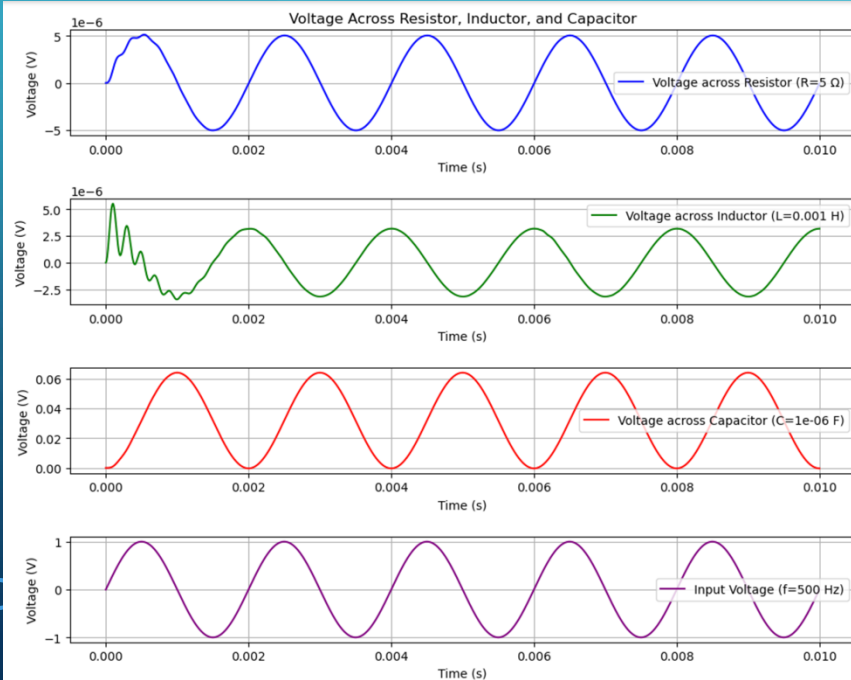


VARYING RESISTANCE

Minimizes distortions in the inductor and stabilizes circuit output.

```
#Varying resistance
frequencies = [500]
resistances = [5, 10, 20] # Varying resistances in Ohms
inductances = [1e-3]
capacitances = [1e-6]

# Loop through different parameter sets and plot the results
for f in frequencies:
    for R in resistances:
        for L in inductances:
            for C in capacitances:
                rlc_response(R, L, C, f)
```

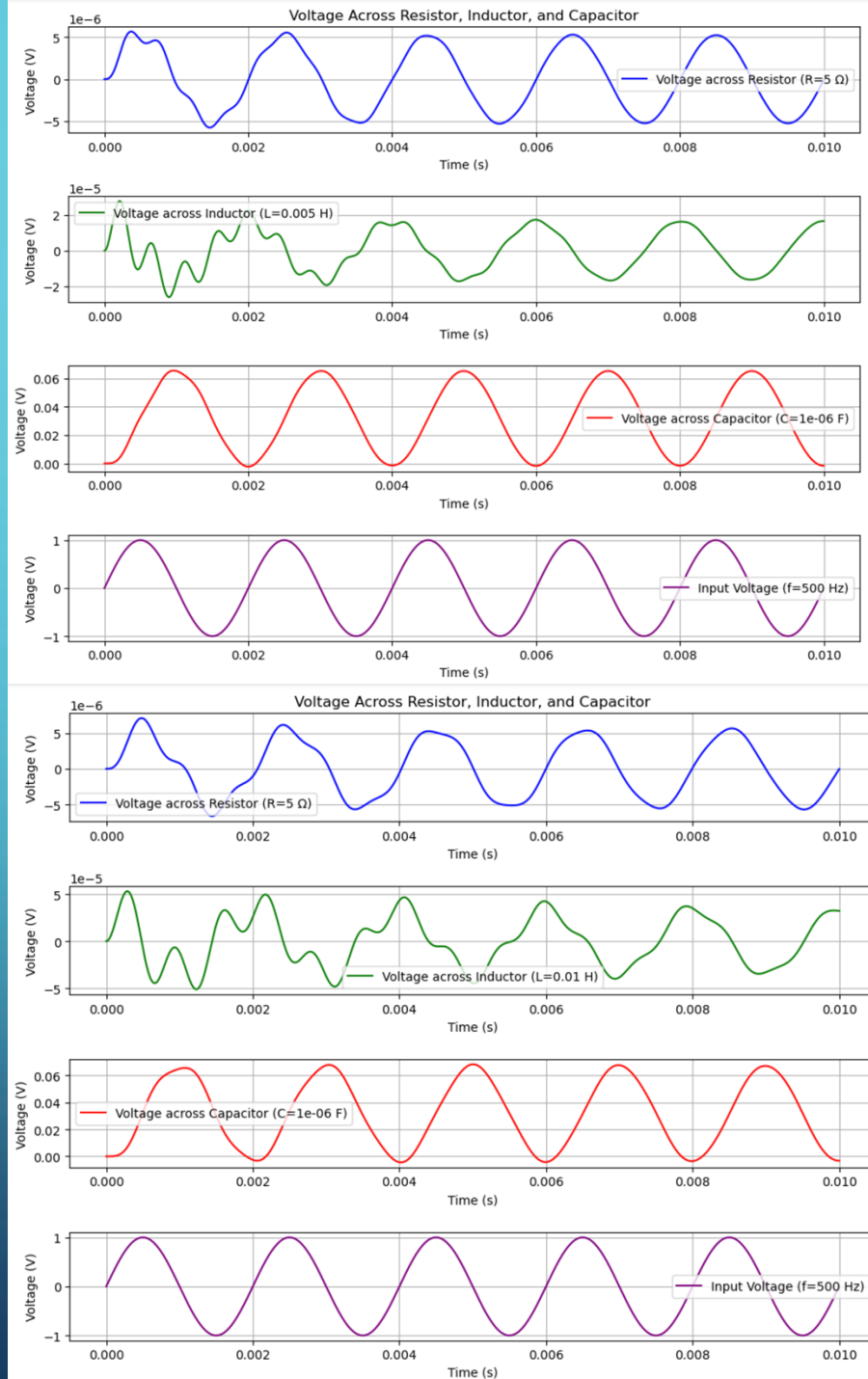
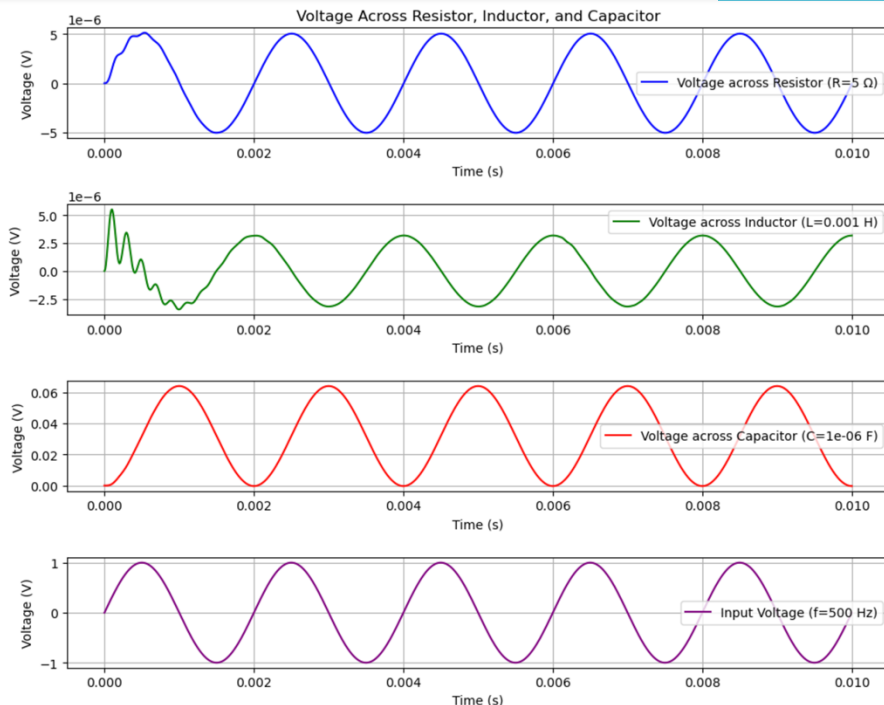


VARYING INDUCTANCE

Increases distortions across all component outputs.

```
#Varying inductance
frequencies = [500]
resistances = [5]
inductances = [1e-3, 5e-3, 1e-2] # Varying inductances in Henrys
capacitances = [1e-6]
```

```
# Loop through different parameter sets and plot the results
for f in frequencies:
    for R in resistances:
        for L in inductances:
            for C in capacitances:
                rlc_response(R, L, C, f)
```



VARYING CAPACITANCE

Stabilizes the circuit but creates straight bumps.

```
#Varying capacitances
frequencies = [500]
resistances = [5]
inductances = [1e-3]
capacitances = [1e-6, 5e-6, 1e-5] # Varying capacitances in Farads

# Loop through different parameter sets and plot the results
for f in frequencies:
    for R in resistances:
        for L in inductances:
            for C in capacitances:
                rlc_response(R, L, C, f)
```

